

MODEL ANSWER (ideal worked solution):

Question 2 (4 marks)

Find the range of values of x for which $(2x - 1)(x + 3) > 0$. Show all working on a number line.

RUBRIC (marking criteria):

MODEL ANSWER (ideal worked solution):

Question 3 (4 marks)

Given that $\log_2(x + 3) + \log_2(x - 1) = 5$, find the value of x .

RUBRIC (marking criteria):

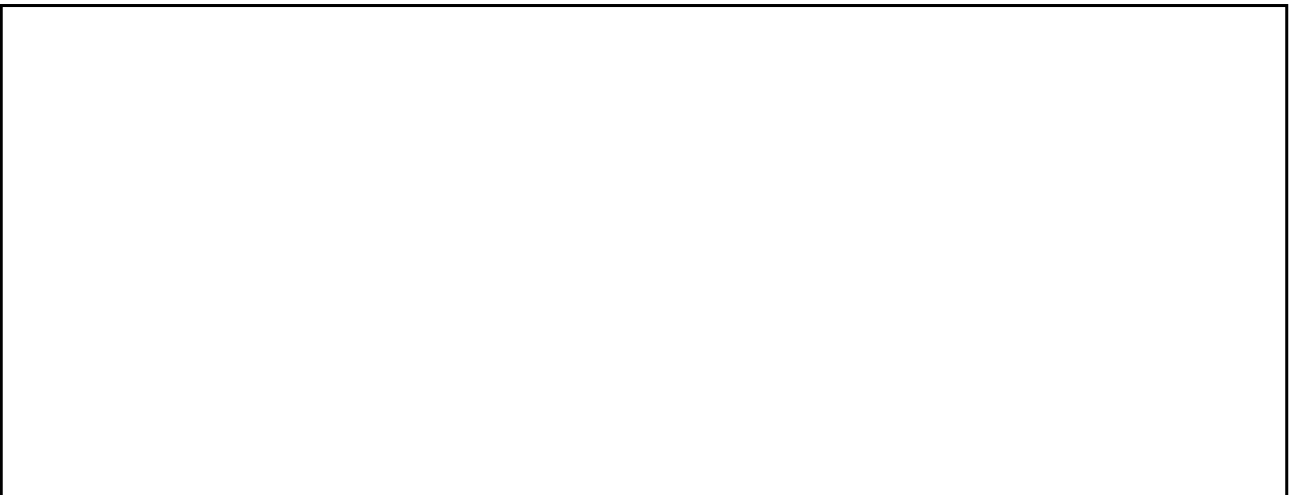
MODEL ANSWER (ideal worked solution):



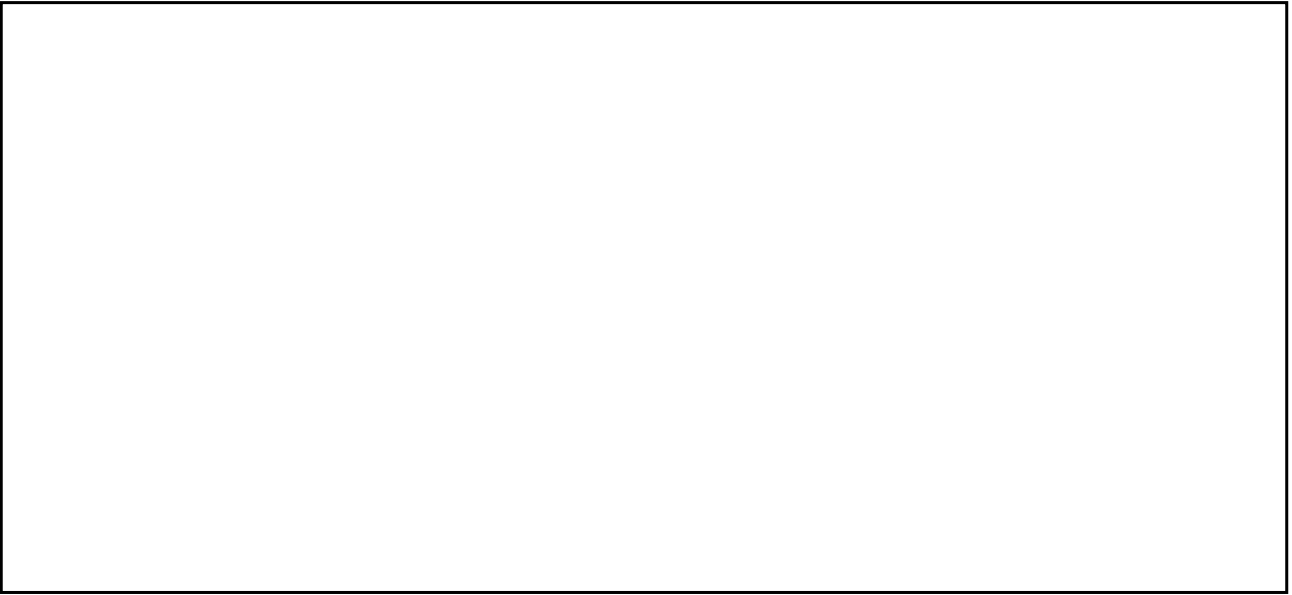
Question 4 (3 marks)

Find the equation of a straight line passing through $(-2, 5)$ and perpendicular to the line $2x - 3y + 6 = 0$.

RUBRIC (marking criteria):



MODEL ANSWER (ideal worked solution):

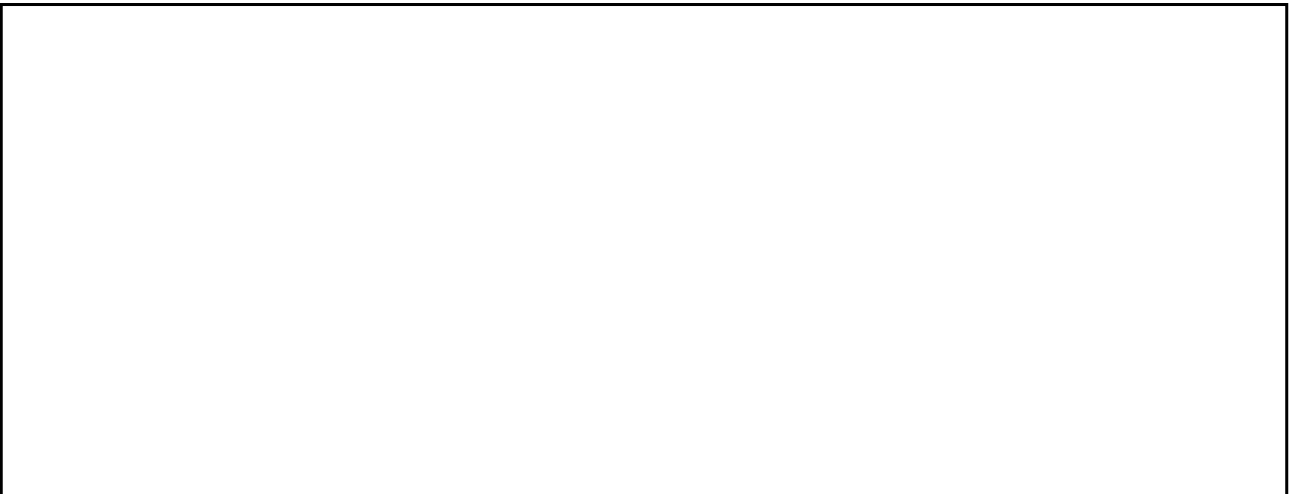


Question 5 (4 marks)

A triangle has vertices A(1, 2), B(5, 2) and C(5, 6). Find:

- (a) The length of AC. [2 marks]

RUBRIC (marking criteria):



MODEL ANSWER (ideal worked solution):



(b) The midpoint M of AC, and show that BM is perpendicular to AC. [2 marks]

RUBRIC (marking criteria):

MODEL ANSWER (ideal worked solution):

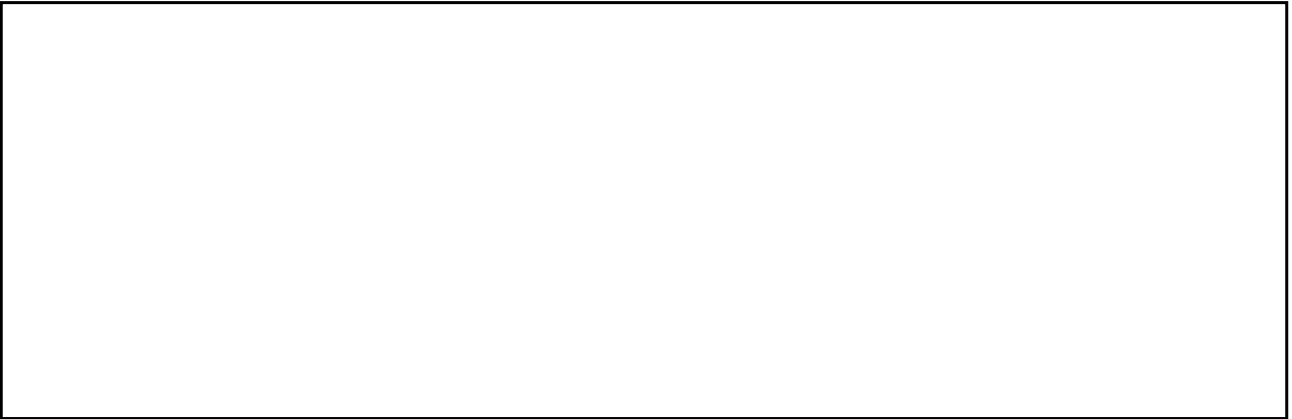
Question 6 (4 marks)

Differentiate with respect to x:

(a) $f(x) = 3x^4 - 5x^2 + 7x - 2$ [2 marks]

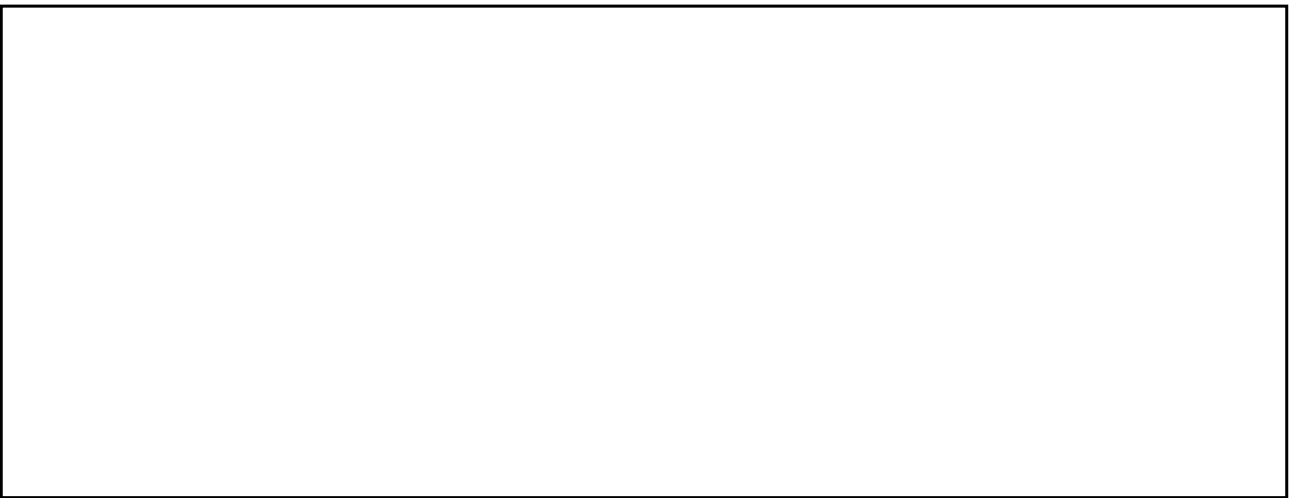
RUBRIC (marking criteria):

MODEL ANSWER (ideal worked solution):

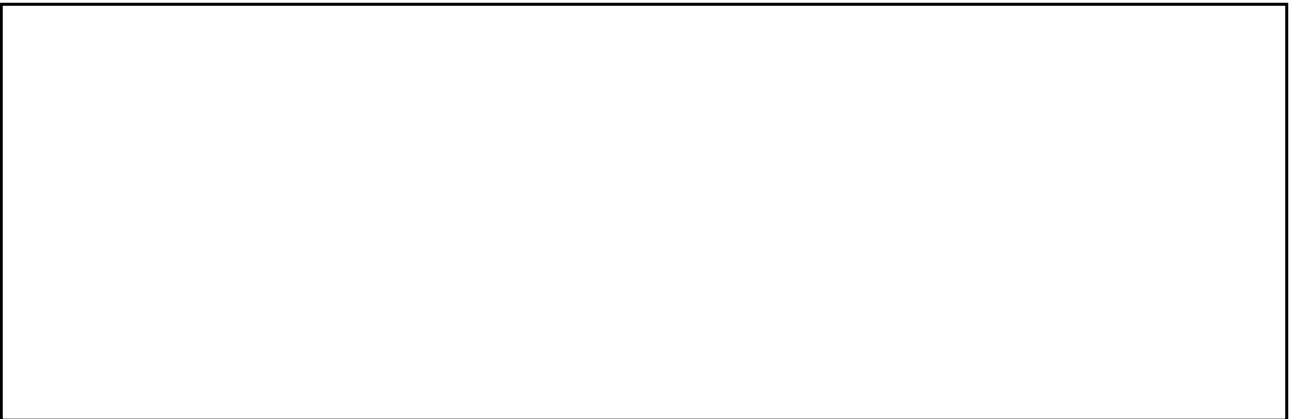
A large, empty rectangular box with a black border, intended for the model answer.

(b) $g(x) = (2x + 1)^3$ [2 marks]

RUBRIC (marking criteria):

A large, empty rectangular box with a black border, intended for the marking criteria.

MODEL ANSWER (ideal worked solution):

A large, empty rectangular box with a black border, intended for the model answer.

Question 7 (4 marks)

The table below shows marks scored by 10 students in a test:

Marks: 12, 18, 15, 20, 14, 19, 17, 13, 16, 21

- (a) Calculate the mean. [1 mark]

RUBRIC (marking criteria):

MODEL ANSWER (ideal worked solution):

- (b) Calculate the standard deviation. [3 marks]

RUBRIC (marking criteria):

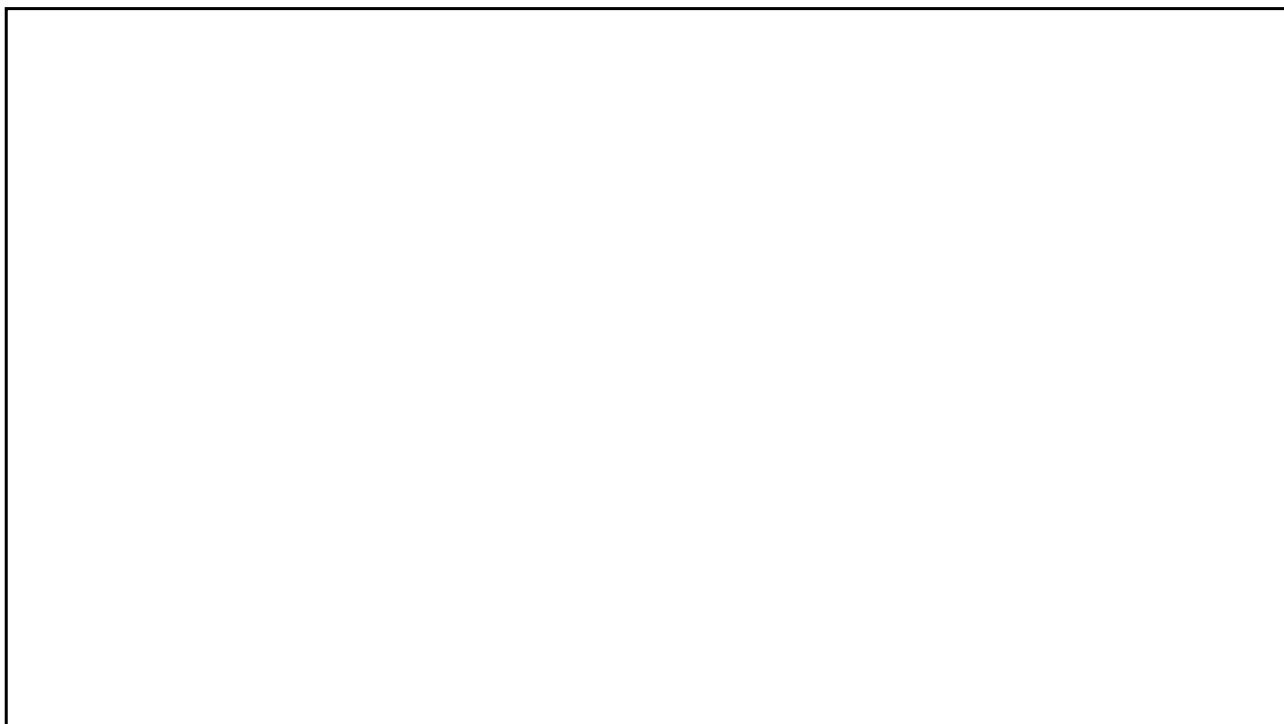
MODEL ANSWER (ideal worked solution):

Question 8 (4 marks)

Solve the equation $2\sin^2(\theta) - \sin(\theta) - 1 = 0$ for $0^\circ \leq \theta \leq 360^\circ$.

RUBRIC (marking criteria):

MODEL ANSWER (ideal worked solution):



--- End of Marking Scheme ---